KPI & Metrics Reference Document

Sales vs. Inventory Dashboard — Power BI Project

Project	GlowMart BI Dashboard — Scenario 1	**Prepared by**	Head of BI & Data Analytics
Data Period	2022 – 2024 (3 Years)	**Tool**	Microsoft Power BI (DAX)
Customers	500 customers · 5 cities	**Transactions**	15,000 orders · 90 products · 4 brands

Section 1 — Data Model & Table Relationships

The Power BI data model is built on four tables connected by key fields. These relationships form the backbone that allows all KPI measures to filter correctly across cities, brands, dates, and customers.

Tables in the Model

Table	Primary Key	Role in Model	Key Columns
Sales (Fact)	SalesID	Central fact table — all transactions	CustomerID, ProductID, OrderDate, Quantity, Unit Price, Discount, Total Sales
Customers (Dimension)	CustomerID	Describes who bought — filters by city, gender, age	CustomerID, City, Region, Gender, Age Group, Signup Date
Products (Dimension)	ProductID	Describes what was bought — filters by brand, cost	ProductID, Brand, Cost Price
Date (Dimension)	Date	Enables all time intelligence in DAX	Date, Year, Month, Quarter, Year-Month, Day of Week, Is Weekend

Relationships to Create in Power BI

From Table · Column	→	To Table · Column	Cardinality	Direction
`Sales[CustomerID]`	→	`Customers[CustomerID]`	Many-to-One (*:1)	Single (→ Customers)
`Sales[ProductID]`	→	`Products[ProductID]`	Many-to-One (*:1)	Single (→ Products)
`Sales[OrderDate]`	→	`Date[Date]`	Many-to-One (*:1)	Single (→ Date)

Note: All three relationships use Single filter direction flowing from the dimension table into the Sales fact table. This is the standard star-schema setup for Power BI.

Section 2 — KPI Reference (12 Measures)

Each KPI below includes: the business question it answers, the standard/threshold values, what high or low figures mean, the arithmetic formula, and the verified DAX measure code.

KPI Summary Overview

#	KPI Name	Category	Threshold (Low)	Threshold (High)	Actual Value (Dataset)
KPI 01	Total Revenue	Core Revenue	See detail below	See detail below	
KPI 02	Total Transactions (Order Count)	Volume	See detail below	See detail below	
KPI 03	Average Order Value (AOV)	Per-Order Efficiency	See detail below	See detail below	
KPI 04	Total Units Sold	Volume	See detail below	See detail below	
KPI 05	Gross Profit	Profitability	See detail below	See detail below	
KPI 06	Gross Profit Margin %	Profitability	See detail below	See detail below	
KPI 07	Revenue by City	Geographic	See detail below	See detail below	
KPI 08	Revenue by Brand	Brand Performance	See detail below	See detail below	
KPI 09	Total Discount Amount	Discount Impact	See detail below	See detail below	
KPI 10	Year-on-Year Revenue Growth %	Growth Trend	See detail below	See detail below	
KPI 11	Transactions per Customer	Customer Behaviour	See detail below	See detail below	
KPI 12	Revenue per Customer	Customer Value	See detail below	See detail below	

Detailed KPI Breakdown

KPI 01 Total Revenue *[Core Revenue]*

Business Question	How much money did GlowMart generate from all sales transactions across all cities and brands?
Formula	`Total Revenue = SUM of (Unit Price × Quantity × (1 − Discount)) for all transactions`
Standard / Threshold	₦600M–₦700M per 3-year period (≈₦200M–₦235M annually) for 500 customers × 30 avg transactions.
If Higher Than Threshold	Business is growing — more customers, higher spend per order, or effective upselling.
If Lower Than Threshold	Revenue decline — possible churn, heavy discounting eroding value, or city-level stockouts reducing transactions.
Actual (Dataset)	

DAX FORMULA (POWER BI)

```
Total Revenue =
    SUMX(
        Sales,
        Sales[Unit Price] * Sales[Quantity] * (1 - Sales[Discount])
    )
```

KPI 02 Total Transactions (Order Count) *[Volume]*

Business Question	How many individual sales orders were placed across all cities and all years?
Formula	`Total Transactions = COUNT of all SalesID records`
Standard / Threshold	15,000 transactions over 3 years (≈5,000/year). Benchmark: 30 transactions per customer on average.
If Higher Than Threshold	High customer engagement and repeat purchase behaviour — healthy for FMCG.
If Lower Than Threshold	Customer inactivity or declining frequency; may signal competitor switching or stockout frustration.
Actual (Dataset)	

DAX FORMULA (POWER BI)

```
Total Transactions =
    COUNTROWS(Sales)
```

KPI 03 Average Order Value (AOV) *[Per-Order Efficiency]*

Business Question	On average, how much revenue does GlowMart earn from a single transaction?
Formula	`AOV = Total Revenue ÷ Total Transactions`
Standard / Threshold	₦40,000–₦50,000 per order (based on 1–5 units × ₦1k–₦20k price range with avg discount ~10%).
If Higher Than Threshold	Customers are buying higher-priced products or more units per order — positive.
If Lower Than Threshold	Orders are small; customers buying minimum quantities or lowest-priced SKUs only.
Actual (Dataset)	

DAX FORMULA (POWER BI)

```
Avg Order Value =
    DIVIDE(
        [Total Revenue],
        [Total Transactions],
        0
    )
```

KPI 04 Total Units Sold *[Volume]*

Business Question	Across all transactions, how many product units were sold in total?
Formula	`Total Units Sold = SUM of Quantity for all Sales rows`
Standard / Threshold	45,000–46,000 units over 3 years. Average of 3 units per transaction (midpoint of 1–5 range).
If Higher Than Threshold	Customers buying in bulk — strong demand, effective promotions.
If Lower Than Threshold	Single-unit purchases dominate; low confidence or affordability issues per visit.
Actual (Dataset)	

DAX FORMULA (POWER BI)

```
Total Units Sold =
    SUM(Sales[Quantity])
```

KPI 05 Gross Profit *[Profitability]*

Business Question	After deducting the cost of products sold, how much profit does GlowMart retain?
Formula	`Gross Profit = Total Revenue − (Cost Price × Quantity Sold)`
Standard / Threshold	GP should be ≥25% of revenue for FMCG beauty retail; industry benchmark is 25%–35%.
If Higher Than Threshold	Strong pricing power, efficient procurement, or low-cost product mix driving margin.
If Lower Than Threshold	Pricing may be too close to cost, or over-discounting is compressing margin below sustainable levels.
Actual (Dataset)	

DAX FORMULA (POWER BI)

```
Gross Profit =
    SUMX(
        Sales,
        (Sales[Unit Price] * Sales[Quantity] * (1 - Sales[Discount]))
        -
        (RELATED(Products[Cost Price]) * Sales[Quantity])
    )
```

KPI 06 Gross Profit Margin % *[Profitability]*

Business Question	What percentage of revenue is retained as gross profit after cost of goods?
Formula	`GP Margin % = (Gross Profit ÷ Total Revenue) × 100`
Standard / Threshold	Target: 25%–35% for FMCG beauty/personal care retail. Minimum acceptable: 20%.
If Higher Than Threshold	Products with high mark-up are leading sales; brand mix is optimal.
If Lower Than Threshold	Excessive discounting or high-cost, low-price products dominate — needs pricing review.
Actual (Dataset)	

DAX FORMULA (POWER BI)

```
GP Margin % =
    DIVIDE(
        [Gross Profit],
        [Total Revenue],
        0
    )
```

KPI 07 Revenue by City *[Geographic]*

Business Question	Which of the 5 cities (Lagos, Abuja, Kano, Ibadan, Port Harcourt) generates the most revenue — and is allocation fair?
Formula	`Revenue by City = Total Revenue filtered by Customer[City]`
Standard / Threshold	Each city should contribute 18%–22% of total revenue if demand is even. Variance >5% flags a disparity.
If Higher Than Threshold	That city has higher demand or better customer density — deserves proportionally more stock.
If Lower Than Threshold	Possibly understocked, lower customer count, or weaker brand preference in that city.
Actual (Dataset)	

DAX FORMULA (POWER BI)

```
Revenue by City =
    CALCULATE(
        [Total Revenue],
        ALLEXCEPT(Customers, Customers[City])
    )

-- Or use as a measure with City on visual axis:
Revenue by City =
    CALCULATE([Total Revenue])
```

KPI 08 Revenue by Brand *[Brand Performance]*

Business Question	Which of the 4 brands (BrandA, B, C, D) is the top revenue contributor — and should receive priority stocking?
Formula	`Revenue by Brand = Total Revenue filtered by Products[Brand]`
Standard / Threshold	In a 4-brand portfolio, each brand should contribute 20%–30%. If one brand exceeds 35%, dependency risk is high.
If Higher Than Threshold	Brand is dominating — prioritise its stock, but review over-reliance risk.
If Lower Than Threshold	Brand is underperforming — may need repositioning, promotion, or discontinuation of low-demand SKUs.
Actual (Dataset)	

DAX FORMULA (POWER BI)

```
Revenue by Brand =
    CALCULATE(
        [Total Revenue],
        ALLEXCEPT(Products, Products[Brand])
    )

-- Brand Revenue Share %:
```

```
Brand Revenue Share % =
    DIVIDE(
        [Total Revenue],
        CALCULATE([Total Revenue], ALL(Products)),
        0
    )
```

KPI 09 Total Discount Amount *[Discount Impact]*

Business Question	How much revenue was surrendered through discounting — and is discounting actually boosting volume?
Formula	`Discount Amount = (Unit Price × Quantity) − Total Sales`
Standard / Threshold	Discount rates of 0%–10% are acceptable for FMCG beauty retail. Above 15% on average indicates promotional dependency.
If Higher Than Threshold	Revenue leakage. Profit is being traded for volume that may not be incremental.
If Lower Than Threshold	Margin is protected. Customers are buying at full or near-full price — strong brand value.
Actual (Dataset)	

DAX FORMULA (POWER BI)

```
Total Discount Amount =
    SUMX(
        Sales,
        (Sales[Unit Price] * Sales[Quantity]) - Sales[Total Sales]
    )

-- Discount Rate %:
Avg Discount Rate % =
    AVERAGE(Sales[Discount])
```

KPI 10 Year-on-Year Revenue Growth % *[Growth Trend]*

Business Question	Is revenue growing, flat, or declining from 2022 to 2023 to 2024?
Formula	`YoY Growth = (Current Year Revenue − Prior Year Revenue) ÷ Prior Year Revenue × 100`
Standard / Threshold	For a growing Nigerian FMCG beauty brand, target ≥5% YoY growth. Flat (0–2%) is a warning. Negative = decline.
If Higher Than Threshold	Business is expanding — marketing, new customers, or improved stock availability is working.
If Lower Than Threshold	Stagnant or declining revenue — urgent strategic review needed across cities and brands.
Actual (Dataset)	

DAX FORMULA (POWER BI)

```
YoY Revenue Growth % =
    VAR CurrentYear = [Total Revenue]
    VAR PriorYear =
        CALCULATE(
            [Total Revenue],
            SAMEPERIODLASTYEAR('Date'[Date])
```

```
        )
    RETURN
    DIVIDE(
        CurrentYear - PriorYear,
        PriorYear,
        BLANK()
    )
```

KPI 11 Transactions per Customer *[Customer Behaviour]*

Business Question	On average, how many times did each customer purchase from GlowMart within the 3-year period?
Formula	`Transactions per Customer = Total Transactions ÷ Total Unique Customers`
Standard / Threshold	30 transactions per customer over 3 years (≈10/year or ~once per month). This is the dataset benchmark.
If Higher Than Threshold	High loyalty and repeat purchase behaviour — strong customer retention.
If Lower Than Threshold	Customers are not returning frequently — churn risk or satisfaction issues.
Actual (Dataset)	

DAX FORMULA (POWER BI)

```
Transactions per Customer =
    DIVIDE(
        [Total Transactions],
        DISTINCTCOUNT(Sales[CustomerID]),
        0
    )
```

KPI 12 Revenue per Customer *[Customer Value]*

Business Question	On average, how much total revenue does each customer contribute over the 3-year period?
Formula	`Revenue per Customer = Total Revenue ÷ Total Unique Customers`
Standard / Threshold	₦1,200,000–₦1,400,000 per customer over 3 years (30 orders × avg ₦43,507 AOV).
If Higher Than Threshold	High-value customers — strong upsell, premium product adoption, or high order frequency.
If Lower Than Threshold	Low customer lifetime contribution — either low frequency or low spend per order.
Actual (Dataset)	

DAX FORMULA (POWER BI)

```
Revenue per Customer =
    DIVIDE(
        [Total Revenue],
        DISTINCTCOUNT(Sales[CustomerID]),
        0
    )
```

Section 3 — Notes on DAX & Model Setup

How to Create Relationships in Power BI

Go to Model View in Power BI Desktop. Drag CustomerID from Sales to Customers, ProductID from Sales to Products, and OrderDate from Sales to Date[Date]. All three should be set as Many-to-One with Single filter direction.

Important DAX Notes

1. RELATED() works because of the active relationships — always verify relationships are active (solid line), not inactive (dashed).
2. CALCULATE() with no filter arguments simply evaluates the measure in the current filter context — this is how city/brand slicers interact with all KPIs.
3. SAMEPERIODLASTYEAR() requires a properly marked Date table. Right-click the Date table in Power BI and select 'Mark as Date Table' with the Date column.
4. DIVIDE(numerator, denominator, 0) is preferred over the / operator to gracefully handle division by zero without breaking visuals.
5. SUMX() iterates row by row — use it whenever you need to calculate per-row expressions (like Unit Price × Quantity × (1-Discount)) before summing.
6. DISTINCTCOUNT(Sales[CustomerID]) counts unique customers who appear in the Sales table — important for Transactions per Customer and Revenue per Customer KPIs.

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